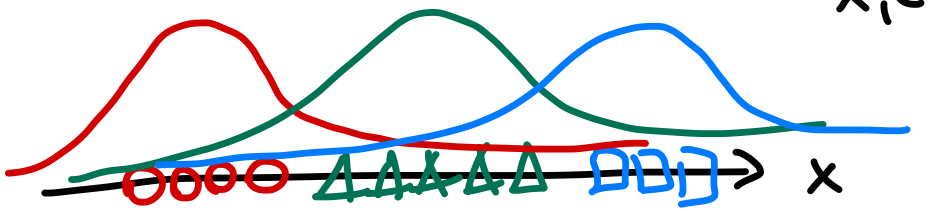
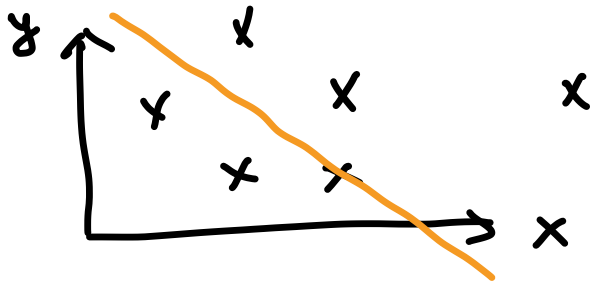
Multivariate Methods



$x_i \in \mathbb{R}^1$



$x_i \in \mathbb{R}^1$

$\Rightarrow$ multiple measurements are taken from each point

$x_i \in \mathbb{R}^D$

$$\mathbf{x}_i = [x_{i1} \; x_{i2} \; \cdots \; x_{iD}]^T$$

x_{i1}: first feature; x_{i2}: second feature; x_{iD}: Dth feature

$y_i \Rightarrow$ target (output) value

$y_i \Rightarrow$ class label

$y_i \in \{1, 2, \ldots, K\}$

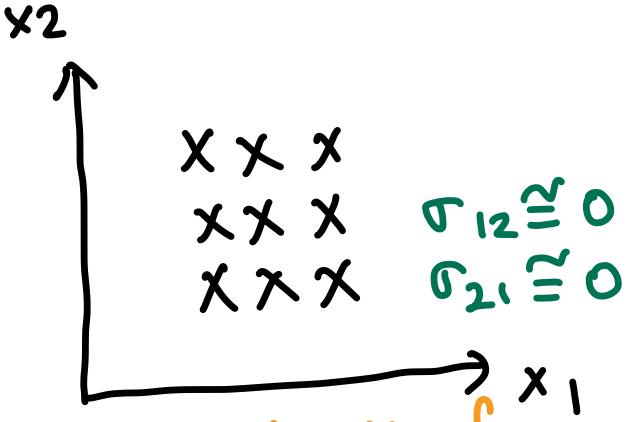
$y_i \in \mathbb{R}$

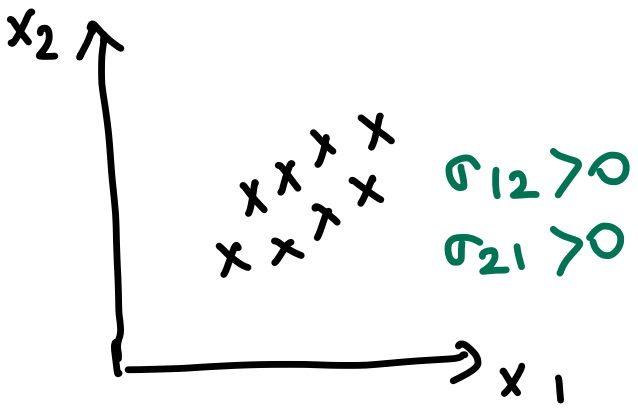
$$\mathcal{X} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N} \qquad \mathbf{x}_i \in \mathbb{R}^D$$

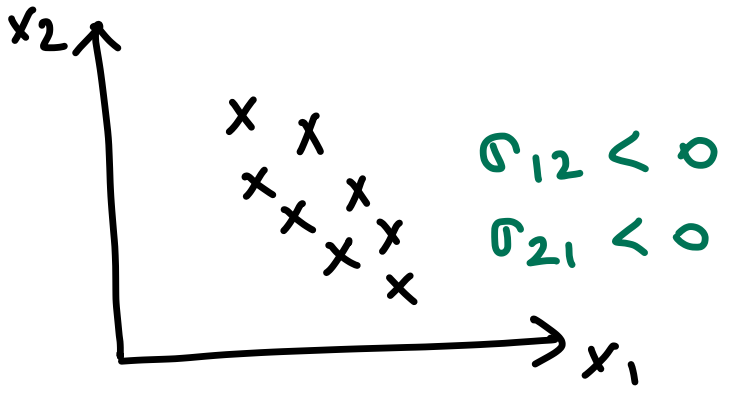
data matrix

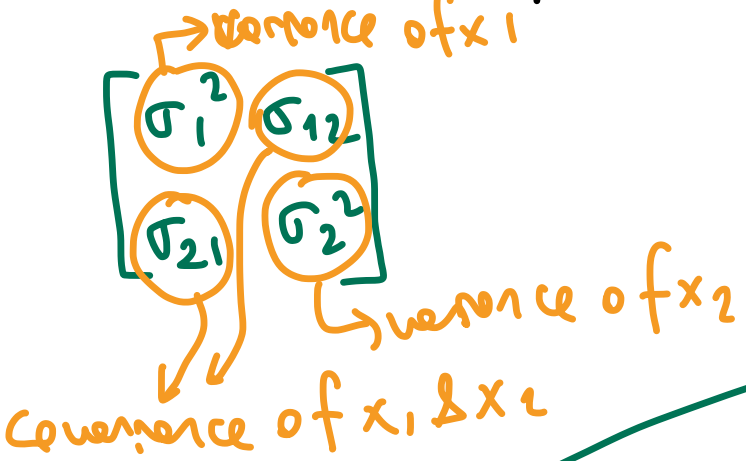
$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1D} \\ x_{21} & x_{22} & \cdots & x_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{ND} \end{bmatrix} \begin{matrix} \rightarrow \mathbf{x}_1^T \\ \rightarrow \mathbf{x}_2^T \\ \\ \rightarrow \mathbf{x}_N^T \end{matrix} \quad N \times D$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}_{N \times 1}$$









Univariate $\Rightarrow$ $x \sim N(x; \mu, \sigma^2)$

multivariate $\Rightarrow$ $\mathbf{x} \sim N(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

mean vector, covariance matrix

$$\hat{\mu}_{1\times 1} = \frac{\sum_{i=1}^{N} x_{i\,1\times 1}}{N}$$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{N} (x_i - \hat{\mu})^2}{N}$$

$$\hat{\boldsymbol{\mu}}_{D\times 1} = \frac{\sum_{i=1}^{N} \mathbf{x}_{i\,D\times 1}}{N}$$

$$\hat{\boldsymbol{\Sigma}}_{D\times D} = \frac{\sum_{i=1}^{N} \overbrace{(\mathbf{x}_i - \hat{\boldsymbol{\mu}})}^{D\times 1} \overbrace{(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^T}^{1\times D}}{N}$$

$$N(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$N(x;\mu,\Sigma) = \frac{1}{\sqrt{(2\pi)^D|\Sigma|}} \cdot \exp\left[-\frac{1}{2}(x-\mu)^T \cdot \Sigma^{-1}(x-\mu)\right]$$

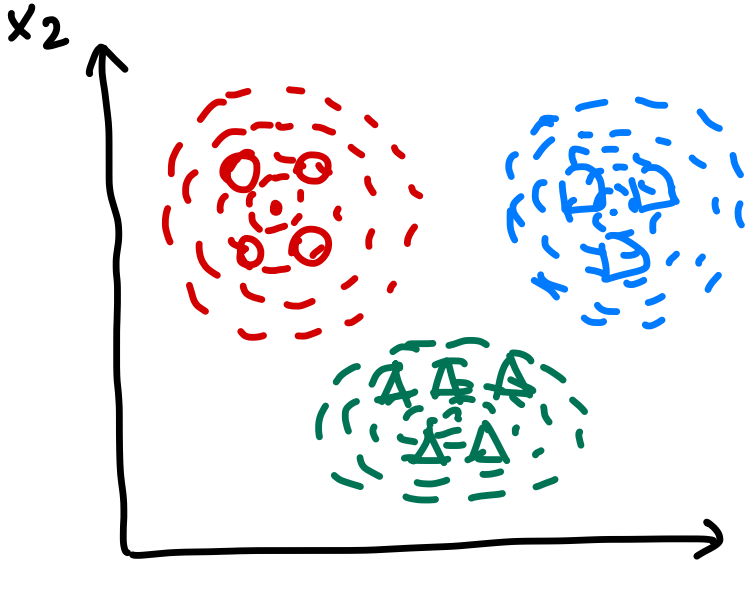
when $D=1$

$$\Sigma = [\sigma^2]$$

$$\frac{1}{\sqrt{(2\pi)^1 \cdot \sigma^2}} \cdot \exp\left[-\frac{1}{2}\overbrace{(x-\mu)^T}^{1\times 1}\frac{1}{\sigma^2}\overbrace{(x-\mu)}^{1\times 1}\right]$$

$$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

Multivariate Parametric Classification



class conditional density

$$p(x|y=c) = N(x; \mu_c, \Sigma_c)$$

$$\frac{1}{\sqrt{(2\pi)^D |\Sigma_c|}} \exp\left[-\frac{1}{2}(x-\mu_c)^T \cdot \Sigma_c^{-1} \cdot (x-\mu_c)\right]$$

$$-\frac{1}{2}(x-a)^T B (x-a) = -\frac{1}{2}x^T \cdot Bx + x^T Ba - \frac{1}{2}a^T Ba$$

$$g_c(x) = \log\left[\hat{p}(x|y=c)\right] + \log\left[\hat{Pr}(y=c)\right]$$

$$g_c(x) = -\frac{D}{2}\log(2\pi) - \frac{1}{2}\cdot\log(|\hat{\Sigma}_c|) - \frac{1}{2}x^T\hat{\Sigma}_c^{-1}\cdot x + x^T\cdot\hat{\Sigma}_c^{-1}\cdot\hat{\mu}_c - \frac{1}{2}\hat{\mu}_c\hat{\Sigma}_c^{-1}\hat{\mu}_c + \log\left[\hat{Pr}(y=c)\right]$$

$$\#\text{ of parameters} = K\left[\frac{D(D+1)}{2} + D\right] + K - 1$$

- K: # of classes
- $\frac{D(D+1)}{2}$: covariance matrix
- D: mean vector
- $K-1$: prior prob.

$a^T \cdot b = b^T a$

$a^T \cdot X \cdot b = b^T \cdot X a$ if X is square & symmetric

$$\hat{\mu}_c = \frac{\sum_{i=1}^{N} [x_i \, 1(y_i = c)]}{\sum_{i=1}^{N} 1(y_i = c)}$$ } N_c = # of datapoints from class c

$$\hat{\Sigma}_c = \frac{\sum_{i=1}^{N} [(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T \cdot 1(y_i = c)]}{\sum_{i=1}^{N} 1(y_i = c)}$$

$$\hat{P}_c(y = c) = \frac{\sum_{i=1}^{N} 1(y_i = c)}{N} = \frac{N_c}{N}$$ } frequency of class c

$ax^2 + bx + c$

$$g_c(x) = x^T \cdot W_c x + w_c^T \cdot x + w_{c0}$$

(uppercase W_c, lowercase w_c)

$$W_c = -\frac{1}{2} \hat{\Sigma}_c^{-1}$$

$$w_c = \hat{\mu}_c^T \cdot \hat{\Sigma}_c^{-1}$$

$$w_{c0} = -\frac{D}{2} \log(2\pi) - \frac{1}{2} \log(|\hat{\Sigma}_c|) - \frac{1}{2} \hat{\mu}_c^T \hat{\Sigma}_c^{-1} \cdot \hat{\mu}_c + \log(\hat{P}_c[y = c])$$